# Zanshin is why you need!

## Attention is syntactic focus;
## Zanshin relaxes this with semantic awareness.

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**Abstract**
Zanshin, attention that has both syntactic and semantic fidelity for natural language processing.

## Introduction

Essentially, Zanshin* learning problems minimize perplexity regularized by average Renyi α-entropy. This achieves both syntactic and semantic fidelity in natural language processing. Noken implicit models can be combined or mutated to produce offspring using genetic algorithms. In the next-token and seq-to-seq Noken learning problems we're about to introduce, the initial target and source Nokens can be such offspring, the fittest resulting from natural selection [3].

Implicit deep learning models [1] consist of a pair of linked equations. The equilibrium equation defines a fixed point state [2] corresponding to the input and the prediction equation output, a function of both input and the fixed point. When the input is a token and the prediction function is constant, we call the fixed point a Noken, a latent token. This form of implicit modeling simultaneously learns key-value and key-query pairs (in addition to Noken) giving semantics to both values and queries, as well as a contextual token representation, given by a weighted sum of values, the weights proportional to the similarity of a query to the value keys. In turn, this contextual representation has a probability distribution over the query keys, that with respect to, we choose the most likely, as the output token. The learning problem is then to minimize the perplexity of output, regularized by average Renyi α-entropy of the distributions over query keys. Perplexity attends to syntactic fidelity while the regularization attends to semantic fidelity. The discussion so far has been in the context of next-token learning. Similar considerations as we shall see also apply to sea-to-seq learning.

Transformer attention is thus superseded by Zanshin learning problems where semantics is now naturally incorporated with syntactic association. The key to this (pun intended) rests on the choice of similarity functions (between keys, queries, and contextual representations). The aim is to avoid hallucinations and have meaningful chats.

* Zanshin is a state of awareness; of relaxed alertness, in Japanese martial arts. A literal translation of zanshin is "remaining mind". So, focused attention to the task at hand, and at the same time, having relaxed situational awareness, like a cat.

## Noken learning problems

Input tokens: target $\{t_i\}_{i=1}^{m} \subset \mathbb{R}^{r}$

source $\{t_i'\}_{i=1}^{n} \subset \mathbb{R}^{r'}$

## Noken implicit models

Learn matrices $V$, $Q$, $\mathcal{N}$, $K$ and fixed points

$s_i \in \mathbb{R}^n$, $n > r$, latent tokens called <u>Noken</u>, so that:

$$s_i = \varphi(V s_i + Q t_i)$$

$$\varepsilon = \mathcal{N} s_i + K t_i$$

where $\varepsilon \in \mathbb{R}^n$.

$V = \Theta - \operatorname{diag}(|\Theta| \mathbb{1}_n) + \delta I_n$ where

$\Theta \in \mathbb{R}^{n \times n}$, $\delta < 1$, $\mathbb{1}_n = \begin{bmatrix} 1 \\ \vdots \\ 1 \end{bmatrix}_{n \times 1}$ and $I_n \in \mathbb{R}^{n \times n}$ identity

ensures $V$ is well-posed for $\varphi$,

ie fixed points $s_i$ exist and are unique.

Given $t_i \in \{t_1, \dots, t_m\}$ define:

$$\alpha_{ij} = \underset{j \in [m]}{\text{softmax}} \; \text{score}(q_i, k_j), \quad q_i = Q t_i, \quad k_j = K s_j$$

score a desired similarity function, like a kernel

$$h_i = \sum_{j=1}^{m} \alpha_{ij} v_j, \quad v_j = V s_j$$

$$\beta_{ij} = \underset{j \in [m]}{\text{softmax}} \; \text{score}(h_i, k_j), \quad k_j = K t_j$$

$$j_i^{\beta} = \underset{j \in [m]}{\arg\max} \; \beta_{ij}$$

$$E(\{\beta_{ij}\}) = \frac{1}{m} \sum_{i=1}^{m} H_{\alpha}^{i}, \quad H_{\alpha}^{i} = H_{\alpha}(\{\beta_{ij}\}_{j=1}^{m})$$

output $o_i = t_{j_i^{\beta}}$.

[ Similarly for $t_i' \in \{t_1', \dots, t_n'\}$

define $\alpha_{ij}'$, contextual representation $h_i'$, $\beta_{ij}'$. ]

<u>Learning problem (next-token)</u> For $\lambda > 0$,

$$\text{minimize} \left\{ \text{Perplexity}(o_1, \dots, o_m) + \lambda E(\{\beta_{ij}\}) \right\}$$

For $\{t_i\}_{i=1}^m$ learn $\{h_i\}_{i=1}^m$ and $\{\beta_{ij}\}$

Similarly, For $\{t_i'\}_{i=1}^n$ learn $\{h_i'\}_{i=1}^n$ and $\{\beta_{ij}'\}$

Given input $t_i$, $i \in [m]$,

define $\gamma_{ij} = \underset{j \in [n]}{\text{softmax}} \; \text{score}(h_i, h_j')$

and let $i' = j_i^\gamma = \underset{j \in [n]}{\arg\max} \; \gamma_{ij} \in [n]$

Define output $O_i = t'_{j^{\beta'}_{i'}}$, the translation of $t_i$.

$$[\; t_i \rightsquigarrow h_i \rightsquigarrow h'_{j_i^\gamma} \rightsquigarrow t'_{j^{\beta'}_{(j_i^\gamma)}} \;]$$

Let $E(\{\gamma_{ij}\}) = \frac{1}{m} \sum_{i=1}^m H_\alpha^i$, $H_\alpha^i = H_\alpha(\{\gamma_{ij}\}_{j=1}^n)$

Learning problem (seq-to-seq) For $\lambda > 0$,

$$\text{minimize} \left\{ \text{Perplexity}(O_1, \ldots, O_m) + \lambda E(\{\gamma_{ij}\}) \right\}$$

—A—

Zanshin!

**References**

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